Mathematics Bootcamp

Introduction to Proofs and Choice Theory

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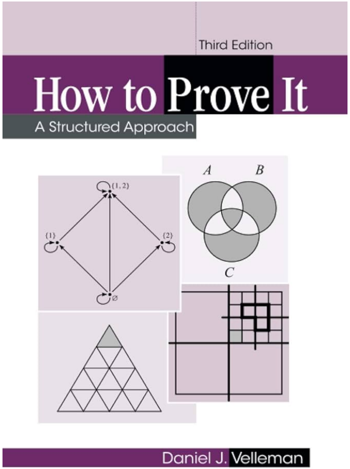
Logic and Precision

- From Ordinary Language to Scientific Language.
- Symbols, Ideas and the Real world.
- Ideas, Truth Value and Obscurantism.

These Slides

- In general, I will use the slides to save time and keep notation for the theorems: straight book screenshots.
- But most of the work will be in the board.
  We learn a language by practice: trial and error.
- Some believe that studying measure theory is a great way to start. This is like syntactic analysis for a baby in his first words.

Just Prove It.



- Mathematical statements have a **truth value**: each is either true or false.
- Logic tells us which conclusions follow from stated assumptions.
- Most of what follows is taken from Velleman's book.

Structured Programming

```
do
    if [condition]
        [List of instructions goes here.]
    else
        [Alternative list of instructions goes here.]
while [condition]
```

Proofs as Structured Programming

Let x be arbitrary.

Suppose $P(x)$ is true.

[Proof of $Q(x)$ goes here.]

Thus, if $P(x)$ then $Q(x)$.

Thus, for all x, if $P(x)$ then $Q(x)$.

Logical Connectives

Conjunction: $P \wedge Q$

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction: $P \vee Q$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Negation: $\neg P$

P	$\neg P$
T	F
F	T

Quantifiers and Negation

Quantifiers

Fix a domain X. A quantifier binds the variable x:

$$\begin{aligned} \forall x \in X\ P(x):&\quad P(x) \text{ holds for every } x \in X, \\ \exists x \in X\ P(x):&\quad P(x) \text{ holds for at least one } x \in X. \end{aligned}$$

An occurrence of x outside the scope of a quantifier is *free*.

Negation laws ($\iff$ means “logically equivalent”)

$$\begin{aligned} \neg(P \wedge Q) &\iff \neg P \vee \neg Q, & \neg(P \vee Q) &\iff \neg P \wedge \neg Q, \\ \neg\big[\exists x \in X\ P(x)\big] &\iff \forall x \in X\ \neg P(x), & \neg\big[\forall x \in X\ P(x)\big] &\iff \exists x \in X\ \neg P(x). \end{aligned}$$

Implication: $P \to Q$

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

$P \to Q$ is false only when P is true and Q is false.

The Contrapositive Is Equivalent

P	Q	$\neg Q$	$\neg P$	$\neg Q \to \neg P$	$P \to Q$
T	T	F	F	T	T
T	F	T	F	F	F
F	T	F	T	T	T
F	F	T	T	T	T

$P \to Q$ and $\neg Q \to \neg P$ have identical truth values. Thus either statement may be proved by proving the other.

Proof by Contradiction

P	Q	$\neg Q$	$P \wedge \neg Q$	$\neg(P \wedge \neg Q)$	$P \to Q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	F	T	T
F	F	T	F	T	T

Method for proving $P \to Q$

Assume P and $\neg Q$. If these assumptions imply a contradiction $\perp$, then $P \wedge \neg Q$ is impossible. Therefore $P \to Q$.

Mathematical Induction

Principle of Mathematical Induction

Let $\mathbb{N} = \{1, 2, 3, \ldots\}$ and fix $n_0 \in \mathbb{N}$. For each natural number $n \geq n_0$, let $P(n)$ be a statement. Suppose that:

1. $P(n_0)$ is true.
2. For every $k \geq n_0$, if $P(k)$ is true, then $P(k+1)$ is true.

Then $P(n)$ is true for every natural number $n \geq n_0$.

From Logic to Set Theory

Definition 1.4.1

The *intersection* of two sets A and B is the set $A \cap B$ defined as follows:

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

The *union* of A and B is the set $A \cup B$ defined as follows:

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}.$$

The *difference* of A and B is the set $A \setminus B$ defined as follows:

$$A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}.$$

Set Theory

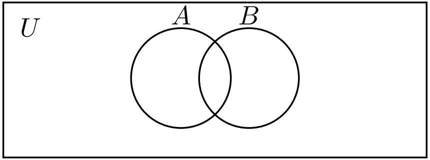


Figure 1.7.

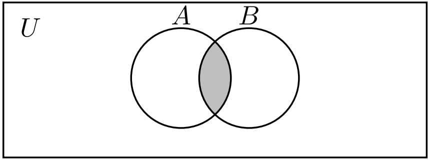


Figure 1.8. $A \cap B$.

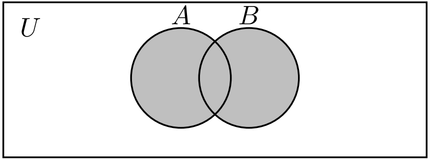


Figure 1.9. $A \cup B$.

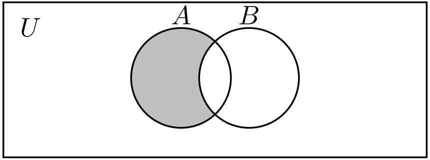


Figure 1.10. $A \setminus B$.

Set Theory

Definition 1.4.5

Suppose A and B are sets. We will say that A is a *subset* of B if every element of A is also an element of B. We write $A \subseteq B$ to mean that A is a subset of B. A and B are said to be *disjoint* if they have no elements in common. Note that this is the same as saying that the set of elements they have in common is the empty set, or in other words $A \cap B = \varnothing$.

- Given $A \subseteq U$, its *complement in* U is

$$A^c = U \setminus A = \{x \in U \mid x \notin A\}.$$

- De Morgan's Laws also apply in Sets:

$$(A \cup B)^c = A^c \cap B^c,$$
$$(A \cap B)^c = A^c \cup B^c.$$

Set Theory

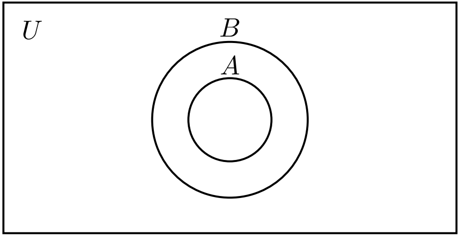


Figure 1.14. $A \subseteq B$.

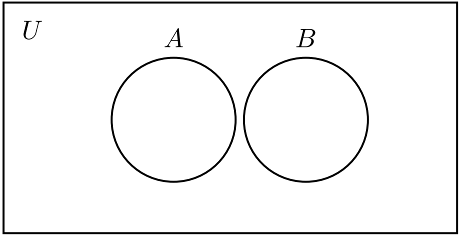


Figure 1.15. $A \cap B = \varnothing$.

Set Theory

Definition 4.1.1

Suppose A and B are sets. Then the *Cartesian product* of A and B, denoted $A \times B$, is the set of all ordered pairs in which the first coordinate is an element of A and the second is an element of B. In other words,

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}.$$

Binary Relations

Definition 4.2.1

Suppose A and B are sets. A set $R \subseteq A \times B$ is a *relation from A to B*. A *relation on A* is a set $R \subseteq A \times A$.

Properties of a Relation on A

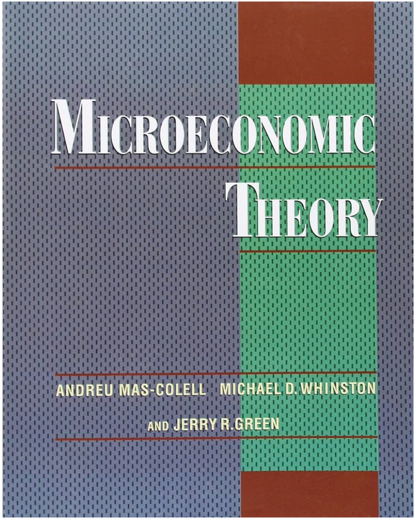
For a relation R on A:

1. *Reflexive*: xRx for every $x \in A$.
2. *Complete*: xRy or yRx (or both) for every $x, y \in A$.
3. *Symmetric*: $xRy \to yRx$ for every $x, y \in A$.
4. *Transitive*: $(xRy \wedge yRz) \to xRz$ for every $x, y, z \in A$.

Time for Economics

- We have seen some mathematical objects and their properties.
- More importantly, these are the words in the formal language that we use in Economics.
- Let's try to read some Economics.

MWG: Microeconomic Theory



- Direct Screenshots:
  Feel that you can learn from it.
- This should not be your unique resource.
- In my 1st year, I also used:
  - **Levin Teaching Notes**
  - Rubinstein Notes Book
  - Kreps Books: 1990, 2012

Individual Decision Making

CHAPTER

1

Preference and Choice

1.A Introduction

In this chapter, we begin our study of the theory of individual decision making by considering it in a completely abstract setting. The remaining chapters in Part I develop the analysis in the context of explicitly economic decisions.

The starting point for any individual decision problem is a *set of possible* (*mutually exclusive*) *alternatives* from which the individual must choose. In the discussion that follows, we denote this set of alternatives abstractly by X.

Preference-Based Approach

1.B Preference Relations

In the preference-based approach, the objectives of the decision maker are summarized in a *preference relation*, which we denote by $\succsim$. Technically, $\succsim$ is a binary relation on the set of alternatives X, allowing the comparison of pairs of alternatives $x, y \in X$. We read $x \succsim y$ as "x is at least as good as y." From $\succsim$, we can derive two other important relations on X:

(i) The *strict preference* relation, $\succ$, defined by

$$x \succ y \iff x \succsim y \text{ but not } y \succsim x$$

and read "x is preferred to y."[1]

(ii) The *indifference* relation, $\sim$, defined by

$$x \sim y \iff x \succsim y \text{ and } y \succsim x$$

and read "x is indifferent to y."

Preference-Based Approach

In much of microeconomic theory, individual preferences are assumed to be *rational*. The hypothesis of rationality is embodied in two basic assumptions about the preference relation $\succsim$: *completeness* and *transitivity*.[2]

Definition 1.B.1: The preference relation $\succsim$ is *rational* if it possesses the following two properties:

(i) *Completeness:* for all $x, y \in X$, we have that $x \succsim y$ or $y \succsim x$ (or both).

(ii) *Transitivity:* For all $x, y, z \in X$, if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

- *Do not waste any attention* speculating about rationality (yet!).
- **Assume** a DM with rational preferences and see where it leads...

This is exactly the approach we follow.

Preference-Based Approach

Proposition 1.B.1: If $\succsim$ is rational then:

(i) $\succ$ is both *irreflexive* ($x \succ x$ never holds) and *transitive* (if $x \succ y$ and $y \succ z$, then $x \succ z$).

(ii) $\sim$ is *reflexive* ($x \sim x$ for all x), *transitive* (if $x \sim y$ and $y \sim z$, then $x \sim z$), and *symmetric* (if $x \sim y$, then $y \sim x$).

(iii) if $x \succ y \succsim z$, then $x \succ z$.

- We can prove this proposition. Let's do it.

Choice-Based Approach

1.C Choice Rules

In the second approach to the theory of decision making, choice behavior itself is taken to be the primitive object of the theory. Formally, choice behavior is represented by means of a *choice structure*. A choice structure $(\mathscr{B}, C(\cdot))$ consists of two ingredients:

(i) $\mathscr{B}$ is a family (a set) of nonempty subsets of X; that is, every element of $\mathscr{B}$ is a set $B \subset X$. By analogy with the consumer theory to be developed in Chapters 2 and 3, we call the elements $B \in \mathscr{B}$ *budget sets*.

(ii) $C(\cdot)$ is a *choice rule* (technically, it is a correspondence) that assigns a nonempty set of chosen elements $C(B) \subset B$ for every budget set $B \in \mathscr{B}$. When $C(B)$ contains a single element, that element is the individual's choice from among the alternatives in B. The set $C(B)$ may, however, contain more than one element. When it does, the elements of $C(B)$ are the alternatives in B that the decision maker *might* choose; that is, they are her *acceptable alternatives* in B. In this case, the set $C(B)$

Choice Behavior Is the Primitive Object

A Choice Rule as a Set-Valued Map

$$C : \mathcal{B} \to 2^X \setminus \{\varnothing\}, \qquad B \mapsto C(B) \subseteq B.$$

- B is the set of alternatives available to the decision maker.
- $C(B)$ is the set of acceptable choices from B; it may contain more than one alternative.
- C has exactly one output for each B, but that output is itself a set.

Order of the Argument

Do not waste any attention speculating about rationality (yet!): first record choices, then ask whether one rational relation explains them.

Choice-Based Approach

Definition 1.C.1: The choice structure $(\mathscr{B}, C(\cdot))$ satisfies the *weak axiom of revealed preference* if the following property holds:

If for some $B \in \mathscr{B}$ with $x, y \in B$ we have $x \in C(B)$, then for any $B' \in \mathscr{B}$ with $x, y \in B'$ and $y \in C(B')$, we must also have $x \in C(B')$.

Definition 1.C.2: Given a choice structure $(\mathscr{B}, C(\cdot))$ the *revealed preference relation* $\succsim^*$ is defined by

$$x \succsim^* y \Leftrightarrow \text{there is some } B \in \mathscr{B} \text{ such that } x, y \in B \text{ and } x \in C(B).$$

WARP Rules Out a Revealed Reversal

Two Observed Menus

$$B = \{a, b\}, \quad C(B) = \{a\}, \qquad B' = \{a, b, c\}, \quad C(B') = \{b\}.$$

Both a and b are available in both menus. We first observe a chosen when b is available; we then observe b chosen while a disappears from the choice set. This violates WARP.

What WARP Does

WARP detects a revealed reversal across menus. It requires no story about the decision maker's motives.

Rationalizing Choice

Rationalization

A preference relation $\succeq$ *rationalizes* a choice rule C on $\mathcal{B}$ if, for every $B \in \mathcal{B}$,

$$C(B) = \{x \in B : x \succeq y \text{ for every } y \in B\}.$$

The choice rule is *rationalizable* if some rational preference relation rationalizes it.

Thus, observed choice contains exactly the best available alternatives according to one complete and transitive relation.

From Choice to Rational Preferences

WARP and Rationalizability

Suppose $\mathcal{B}$ contains every nonempty subset of X with at most three alternatives, and $C(B) \neq \varnothing$ for every $B \in \mathcal{B}$. Then C is rationalizable if and only if it satisfies WARP.

- Menus with two alternatives recover pairwise revealed comparisons.
- Menus with three alternatives test whether those comparisons can be transitive.

Scope

This equivalence uses a rich domain of menus. Observed price-income budgets require a different revealed-preference theorem.

Choice, Preferences, and Utility Are Distinct

- **Choice $\rightarrow$ preferences:** Can one complete and transitive relation reproduce every observed choice set?
- **Preferences $\rightarrow$ utility:** Can those rankings be represented by real numbers?
- **Preferences and a menu $\rightarrow$ choice:** Does the menu contain a best alternative?

Different Questions, Different Assumptions

WARP and a rich menu domain address rationalization. Days 2 and 3 study utility representation. Existence of a best alternative requires additional assumptions on the menu and the representation.